

Geometry & mensuration

LENGTHS AREAS VOLUMES

A: area P/C: perimeter/circumference V: volume S: total surface area L: lateral/curved area (arc length in circle section) B, p: base area, perimeter

Plane figures



Square
 $A = a^2, \quad P = 4a, \quad d = a\sqrt{2}$

Rectangle
 $A = lw, \quad P = 2(l + w)$
 $d = \sqrt{l^2 + w^2}$

Parallelogram
 $A = bh = ab \sin \theta, \quad P = 2(a + b)$
 θ is the included angle of sides a, b .

Trapezium / trapezoid
 $A = \frac{1}{2}(a + b)h, \quad P = a + b + c + d$
 a, b are parallel; h is their separation.

Rhombus
 $A = \frac{1}{2}d_1d_2 = ah, \quad P = 4a$

Kite
 $A = \frac{1}{2}d_1d_2, \quad P = 2(a + b)$
 a, b are the two distinct side lengths.

Regular n-gon
 $P = na, \quad A = \frac{1}{2}Pq = \frac{na^2}{4 \tan(\pi/n)}$
 q is the apothem; a is one side.

Ellipse
 $A = \pi ab$
 a, b are semiaxes. No elementary perimeter formula.

Triangle area & centres

Any triangle
 $A = \frac{1}{2}bh = \frac{1}{2}ab \sin C$
 $s = \frac{1}{2}(a + b + c)$
 $A = \sqrt{s(s-a)(s-b)(s-c)}$
Sides a, b, c face angles A_0, B_0, C ; C is between a, b .

Inradius & circumradius
 $A = r_{\text{in}}s = \frac{abc}{4R}$

Equilateral triangle
 $h = \frac{\sqrt{3}}{2}a, \quad A = \frac{\sqrt{3}}{4}a^2$
 $r_{\text{in}} = \frac{\sqrt{3}}{6}a, \quad R = \frac{\sqrt{3}}{3}a$

Circles, arcs & sectors



Circle & annulus
 $d = 2r, \quad C = 2\pi r, \quad A = \pi r^2$
 $A_{\text{annulus}} = \pi(R^2 - r^2) \quad (R \geq r)$

Sector - radians
 $L = r\theta, \quad A = \frac{1}{2}r^2\theta = \frac{1}{2}rL$
 $P_{\text{sector}} = 2r + L \quad (0 < \theta < 2\pi)$

Sector - degrees
 $L = \frac{\theta_{\text{deg}}}{360} 2\pi r$
 $A = \frac{\theta_{\text{deg}}}{360} \pi r^2$

Chord & minor segment
 $c = 2r \sin(\theta/2)$
 $A_{\text{segment}} = \frac{1}{2}r^2(\theta - \sin \theta)$
Use radians, $0 \leq \theta \leq \pi$; segment = sector minus triangle.

Semicircle
 $A = \frac{1}{2}\pi r^2, \quad P = \pi r + 2r$

Angles & trigonometry

Angle conversion
 $180^\circ = \pi \text{ rad}$
 $\theta_{\text{rad}} = \theta_{\text{deg}} \pi / 180$

Right triangle
 $c^2 = a^2 + b^2$
 $\sin \theta = \frac{\text{opp}}{\text{hyp}}, \quad \cos \theta = \frac{\text{adj}}{\text{hyp}}$
 $\tan \theta = \frac{\text{opp}}{\text{adj}} = \frac{\sin \theta}{\cos \theta}$

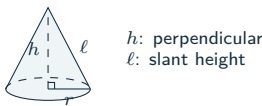
Sine & cosine rules
 $\frac{a}{\sin A_0} = \frac{b}{\sin B_0} = \frac{c}{\sin C} = 2R$
 $c^2 = a^2 + b^2 - 2ab \cos C$

Special right triangles
 $45^\circ : 1 : 1 : \sqrt{2}$
 $30^\circ, 60^\circ, 90^\circ : 1 : \sqrt{3} : 2$
Side ratios in increasing angle order.

Polygon angles
Interior sum = $(n - 2)180^\circ$
Regular interior = $180^\circ - 360^\circ/n$
Regular exterior = $360^\circ/n$

Circle angle facts
Central angle = $2 \times$ inscribed angle
Same arc. An angle in a semicircle is 90° . Opposite angles of a cyclic quadrilateral sum to 180° .

Prisms, cylinders & cones



Cube
 $V = a^3, \quad S = 6a^2, \quad L = 4a^2$
Space diagonal = $a\sqrt{3}$
Cuboid / rectangular prism
 $V = lwh, \quad S = 2(lw + lh + wh)$
 $L = 2h(l + w), \quad d = \sqrt{l^2 + w^2 + h^2}$

Right prism
 $V = Bh, \quad L = ph, \quad S = ph + 2B$

Right circular cylinder
 $V = \pi r^2 h, \quad L = 2\pi r h$
 $S = 2\pi r(h + r)$
Open at one end: $S = 2\pi r h + \pi r^2$.

Right circular cone
 $\ell = \sqrt{r^2 + h^2}$
 $V = \frac{1}{3}\pi r^2 h, \quad L = \pi r \ell$
 $S = \pi r(\ell + r)$

Pyramid
 $V = \frac{1}{3}Bh$
Regular pyramid: $L = \frac{1}{2}pl, \quad S = B + L$
 ℓ is a face slant height, not a sloping edge. Otherwise add the areas of all side faces.

Right conical frustum
 $\ell = \sqrt{h^2 + (R - r)^2}$
 $V = \frac{\pi h}{3}(R^2 + Rr + r^2)$
 $L = \pi(R + r)\ell$
 $S = L + \pi R^2 + \pi r^2$

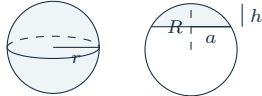
Similar-base pyramidal frustum
 $V = \frac{h}{3}(B_1 + B_2 + \sqrt{B_1 B_2})$
Parallel bases; h is their perpendicular separation.

Surface & volume checks

Perpendicular height
Use h for volume. Use ℓ for the curved area of a cone or the side faces of a regular pyramid.

Composite shapes
Add or subtract component areas and volumes. For exposed surface area, remove joined faces and include exposed cavity walls.

Spheres & curved solids



Sphere
 $V = \frac{4}{3}\pi r^3, \quad S = 4\pi r^2$
Hemisphere
 $V = \frac{2}{3}\pi r^3, \quad L = 2\pi r^2$
 $S = 3\pi r^2$ (including flat base)

Spherical cap
 $V = \frac{\pi h^2}{3}(3R - h), \quad L = 2\pi R h$
 $a^2 = 2Rh - h^2, \quad S = L + \pi a^2$
Sphere radius R ; cap height h ; base radius a ; $0 \leq h \leq 2R$.

Ring torus
 $V = 2\pi^2 R r^2, \quad S = 4\pi^2 R r$
 $R > r$: R is centre-to-tube-centre radius; r is tube radius.

Coordinate geometry

Distance & midpoint
 $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
 $M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$

Straight lines
 $m = \frac{y_2 - y_1}{x_2 - x_1}, \quad y - y_1 = m(x - x_1)$
Parallel: $m_1 = m_2$; $\perp$: $m_1 m_2 = -1$
Slope formulas assume finite slopes; vertical lines are perpendicular to horizontal lines.

Circle & point-to-line distance
 $(x - u)^2 + (y - v)^2 = r^2$
 $d = \frac{|\alpha x_0 + \beta y_0 + \gamma|}{\sqrt{\alpha^2 + \beta^2}}$
Point (x_0, y_0) ; line $\alpha x + \beta y + \gamma = 0$; $(\alpha, \beta) \neq (0, 0)$.

Triangle from three vertices
 $A = \frac{1}{2}|x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$

Similarity & units

Similar figures / solids
Length k , area k^2 , volume k^3
 k is new length / original length.

Convert before substituting
 $1 \text{ m}^2 = 10^4 \text{ cm}^2$
 $1 \text{ m}^3 = 10^6 \text{ cm}^3 = 10^3 \text{ L}$
 $1 \text{ cm}^3 = 1 \text{ mL}$
Lengths: units; areas: units²; volumes: units³. Keep π exact until the final rounding.